MA1521 L1 Week 09 Mon 20261012

Change of Variables in Integration ("by Substitution")

$(x,y) \in \mathbb{R}^2$ —Φ→ $(u,v) \in \mathbb{R}^2$ —$\mathbb{R}$-valued fn f of u,v→ $\mathbb{R}$

original region D = $\Phi^{-1}(\Phi(D))$; image region $\Phi(D)$

specify u = fn of (x,y), v = fn of (x,y)

} require Φ to be invertible, i.e. Φ^{-1} exists

specify x = fn of (u,v), y = fn of (u,v)

Want to evaluate $\iint_{\Phi(D)} f(u,v)\,du\,dv$

composite fn of Φ followed by f

(1st common form)

* Ch. of Var formula: we have

$$\iint_{\Phi(D)} f(u,v)\,du\,dv = \iint_D f(\Phi(x,y))\,\left|J_\Phi(x,y)\right|^{-1}\,dx\,dy$$

$$\iint_{\Phi(D)} f(u,v)\,\left|J_\Phi(\Phi^{-1}(u,v))\right|^{-1}\,du\,dv = \iint_D f(\Phi(x,y))\,dx\,dy = \iint_D f(\Phi(x,y))\,\left|J_{\Phi^{-1}}(\Phi(x,y))\right|^{+1}\,dx\,dy$$

Jacobian determinant

abs val. of

Jacobian determinant of Φ^{-1}

$$J_\Phi(x,y) := \det\begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} \quad \text{fn of } (x,y)$$

$$J_{\Phi^{-1}}(u,v) := \det\begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} \quad \text{fn of } (u,v)$$

Fact: $J_\Phi(\Phi^{-1}(u,v)) = 1 \,/\, J_{\Phi^{-1}}(u,v)$, $J_\Phi(x,y) = 1 \,/\, J_{\Phi^{-1}}(\Phi(x,y))$

(eg) Φ is a linear map, i.e. $u = ax + by$ $\quad (a, b, c, d \in \mathbb{R}$ const.$)$

$v = cx + dy$

i.e. $\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

Then $J_\Phi(x,y) := \det \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$

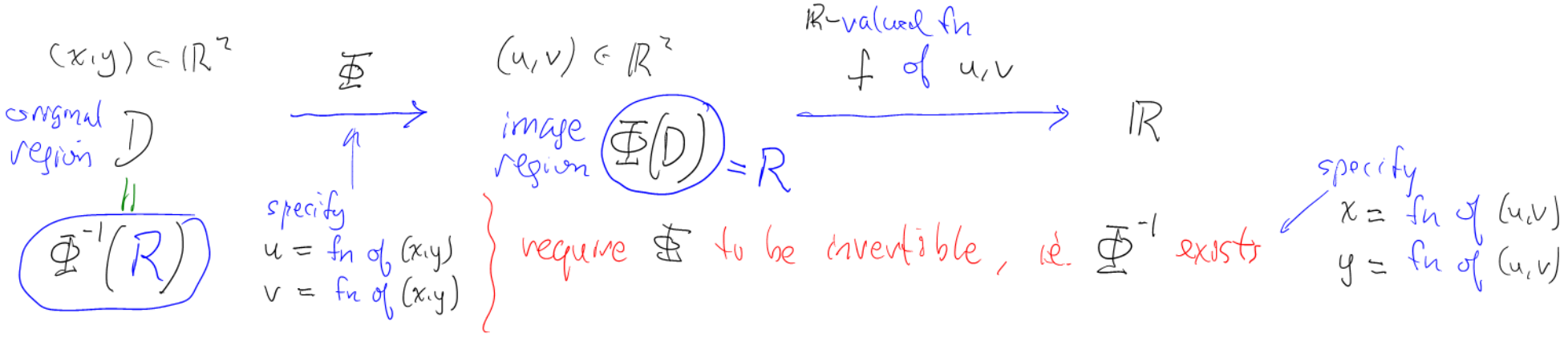
const. fn of (x,y)

↑ special nature when Φ is linear

so Ch. of Var formula simplifies to

$$\iint_{\Phi(D)=R} f(u,v)\, du\, dv = \iint_{D=\Phi^{-1}(R)} f(\Phi(x,y)) \left| ad - bc \right|^{-1} dx\, dy$$

$$= \iint_{D=\Phi^{-1}(R)} f(\Phi(x,y)) \left| \frac{1}{ad-bc} \right| dx\, dy$$



R is region in uv-plane

Want to evaluate $\iint_R f(u,v)\,du\,dv$

(2nd common form)

Ch. of Var formula: we have

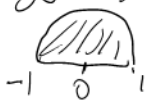
$$\iint_R f(u,v)\,du\,dv = \iint_{\Phi^{-1}(R)} f(\Phi(x,y)) \left| J_\Phi(x,y) \right|^{+1} dx\,dy$$

$$= \iint_{\Phi^{-1}(R)} f(\Phi(x,y)) \left| J_{\Phi^{-1}}(\Phi(x,y)) \right|^{-1} dx\,dy$$

abs val. of

Jacobian determinant of Φ^{-1}

eg polar coordinates eg $\iint_{\text{unit disk}} \frac{1}{\sqrt{1-x^2-y^2}}\,dx\,dy = \int_{\theta=0}^{\theta=\pi}\int_{r=0}^{r=1} \frac{1}{\sqrt{1-r^2}}\, r\,dr\,d\theta$



$\Phi^{-1}(R)$ in (r,θ) plane $\xrightarrow{\Phi}$ $R =$  in xy-plane $\xrightarrow{f(x,y) = \frac{1}{\sqrt{1-x^2-y^2}}}$ $\mathbb{R}$

polar coord Φ

$x = r\cos\theta$
$y = r\sin\theta$

Ch. of Var formula:

$$\iint_R f(x,y)\,dx\,dy = \iint_{\Phi^{-1}(R)} f(\Phi(r,\theta))\left|J_\Phi(r,\theta)\right|^{+1} dr\,d\theta$$

$$= \iint_{\Phi^{-1}(R)} f(\Phi(r,\theta))\left|J_{\Phi^{-1}}(\Phi(r,\theta))\right|^{-1} dr\,d\theta$$

explains why we put $r\,dr\,d\theta$ for polar coords in place of $dx\,dy$

Here Φ specifies $x = r\cos\theta$, $y = r\sin\theta$ in terms of r,θ

so $J_\Phi(r,\theta) = \det\begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \det\begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} = r$ not const

How to remember the formula:

$$D \text{ in } xy\text{-plane} \xrightarrow{\Phi} \Phi(D) \text{ in } uv\text{-plane} \xrightarrow{f} \mathbb{R}$$

$$\Phi^{-1}(R) \quad \xleftarrow{\Phi^{-1}} \quad R \text{ in } uv\text{-plane}$$

(1st common form)

* Ch. of Var formula: we have $\iint_{\Phi(D)} f(u,v)\,du\,dv = \iint_D f(\Phi(x,y))\left|J_\Phi(x,y)\right|^{-1} dx\,dy$ ✗

(2nd common form)

$\iint_{\Phi(D)} f(u,v)\left|J_\Phi(\Phi^{-1}(u,v))\right|^{-1} du\,dv = \iint_D f(\Phi(x,y))\,dx\,dy$ ✓

Ch. of Var formula: we have $\iint_R f(u,v)\,du\,dv = \iint_{\Phi^{-1}(R)} f(\Phi(x,y))\left|J_\Phi(x,y)\right|^{+1} dx\,dy$ ✓

* Think about area: i.e. take f to be const fn 1, and take $\Phi = \begin{pmatrix}\lambda & \\ & \lambda\end{pmatrix}$ to be scaling by λ

then $\iint_R 1\,du\,dv$ is Area(R) $\Rightarrow$ J_Φ is const λ^2

$$= \iint_{\Phi^{-1}(R)} \underbrace{\left|J_\Phi(x,y)\right|^{\pm 1}}_{\lambda^2 \text{ const.}}\, dx\,dy \quad (\pm 1 \overset{?}{})$$

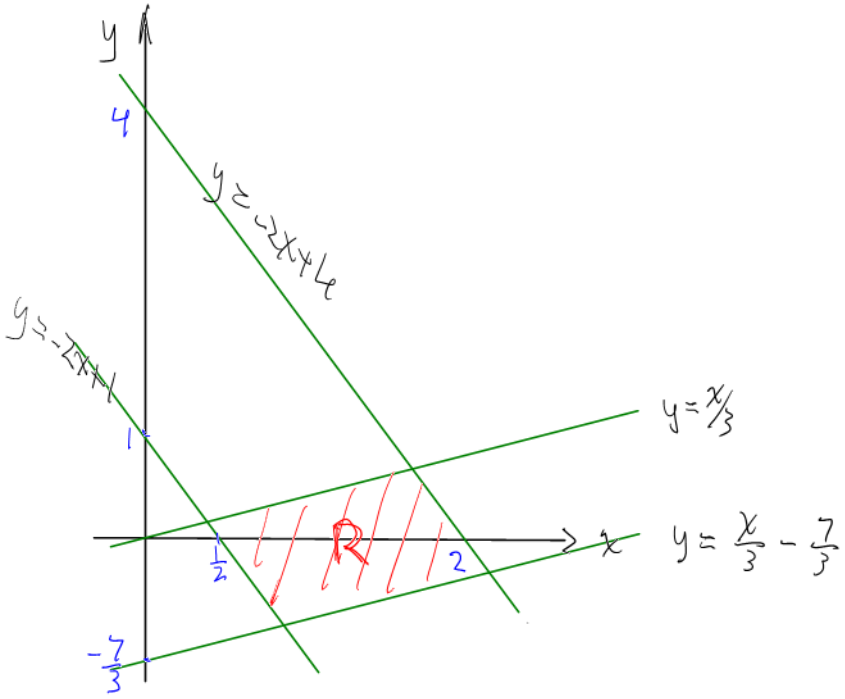
$$= \lambda^{\pm 2 \overset{?}{}} \cdot \underbrace{\text{Area}(\Phi^{-1}(R))}_{\text{will be } \frac{1}{\lambda^2}\cdot \text{Area}(R)}$$

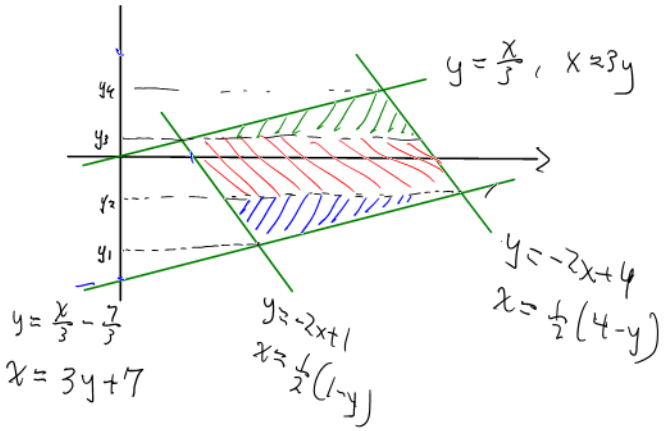
this forces ± 1 to be $+1$

Evaluate $\displaystyle\iint_R \frac{x-3y}{2x+y}\,dx\,dy$

R is the parallelogram bounded on the sides by
$y=-2x+1$ and $y=-2x+4$,
and above and below by
$y=x/3$ and $y=(x-7)/3$.

Use the change of variables $u=x-3y$, $v=2x+y$.





$$\iint_R \frac{x-3y}{2x+y}\,dx\,dy = \int_{y=y_1}^{y=y_2}\int_{x=\frac{1}{2}(1-y)}^{x=3y+7} \frac{x-3y}{2x+y}\,dx\,dy$$



$$+ \int_{y=y_2}^{y=y_3}\int_{x=\frac{1}{2}(1-y)}^{x=\frac{1}{2}(4-y)} \frac{x-3y}{2x+y}\,dx\,dy$$



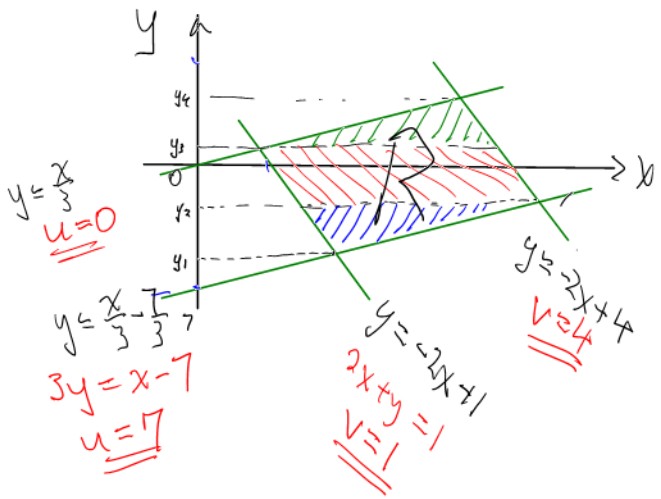
$$+ \int_{y=y_3}^{y=y_4}\int_{x=3y}^{x=\frac{1}{2}(4-y)} \frac{x-3y}{2x+y}\,dx\,dy$$



Super - complicated !!

Remedy: Change Variables

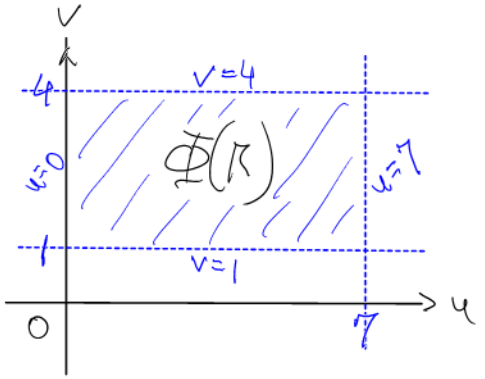
$$u = x - 3y$$
$$v = 2x + y$$



Φ

$$u = x - 3y$$
$$v = 2x + y$$

linear map of (x,y)



$f(u,v) \to \mathbb{R}$

$\frac{u}{v}$

so that

$$f(\Phi(x,y)) = f(x-3y,\ 2x+y) = \frac{x-3y}{2x+y}$$

Ch. of Var

$$\iint_R \frac{x-3y}{2x+y}\,dx\,dy = \iint_{\Phi(R)} \underbrace{\frac{u}{v}}_{f(u,v)} \left| J_\Phi(\Phi^{-1}(u,v)) \right|^{-1} du\,dv$$

$f(\Phi(x,y))$

$$J_\Phi(x,y) = \det\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \det\begin{pmatrix} 1 & -3 \\ 2 & 1 \end{pmatrix} = 7$$ const. indep of (x,y)

so

$$\iint_R \frac{x-3y}{2x+y}\,dx\,dy = \int_{v=1}^{v=4} \int_{u=0}^{u=7} \frac{u}{7v}\,du\,dv$$

$$= \int_{v=1}^{v=4} \frac{1}{7v} \left[\frac{u^2}{2}\right]_{u=0}^{u=7} dv$$

$$= \int_{v=1}^{v=4} \frac{1}{7v} \cdot \frac{49}{2}\, dv = \frac{7}{2} \int_{v=1}^{v=4} \frac{1}{v}\, dv$$

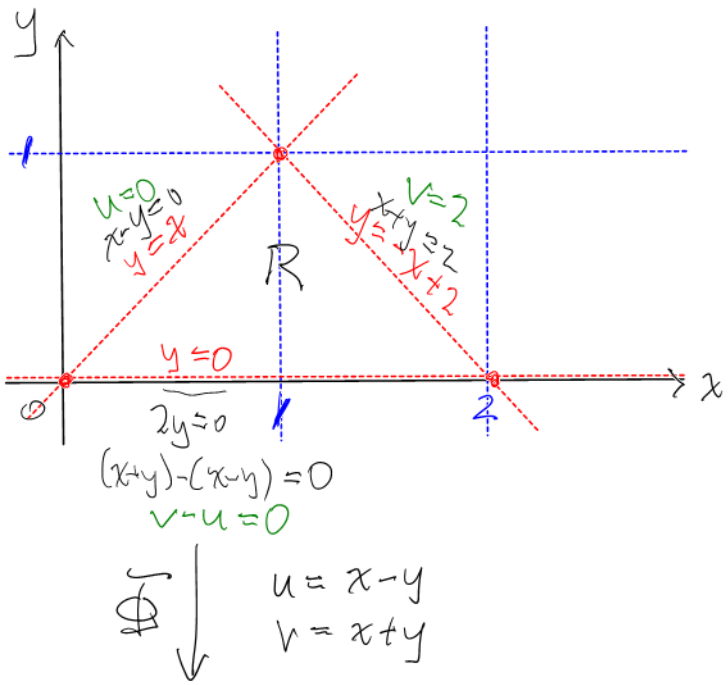
$$= \frac{7}{2} \left[\ln v\right]_{v=1}^{v=4} = \frac{7}{2} \ln 4$$

$$= 7 \ln 2$$

#

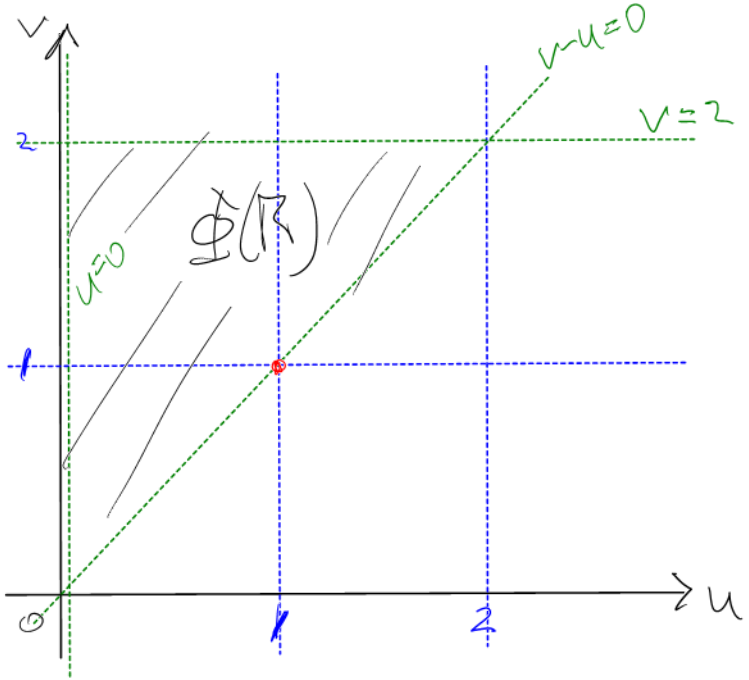
Evaluate $\iint_R \cos\left(\frac{x-y}{x+y}\right)\,dx\,dy$

R is the triangle with vertices at the origin, (2,0) and (1,1).



Funct$_n$ $\cos\left(\frac{x-y}{x+y}\right)$ is complicated as integrand of (x,y).

Use Ch. of Var. $u = x-y$, $v = x+y$



Ch. of Var. formula gives

$$\iint_R \cos\left(\frac{x-y}{x+y}\right) dx\,dy$$

$$= \iint_{\Phi(R)} \cos\frac{u}{v} \left| J_{\Phi}\left(\Phi^{-1}(u,v)\right)\right|^{-1} du\,dv$$

$$= \iint_{\Phi(R)} \cos\frac{u}{v} \times \frac{1}{2}\, du\,dv$$

$$= \int_{v=0}^{v=2} \int_{u=0}^{u=v} \frac{1}{2}\cos\frac{u}{v}\, du\,dv$$

$$= \frac{1}{2}\int_{v=0}^{v=2} \left[v \sin\frac{u}{v}\right]_{u=0}^{u=v} dv$$

$$= \frac{1}{2}\sin 1 \int_{v=0}^{v=2} v\, dv = \sin 1$$

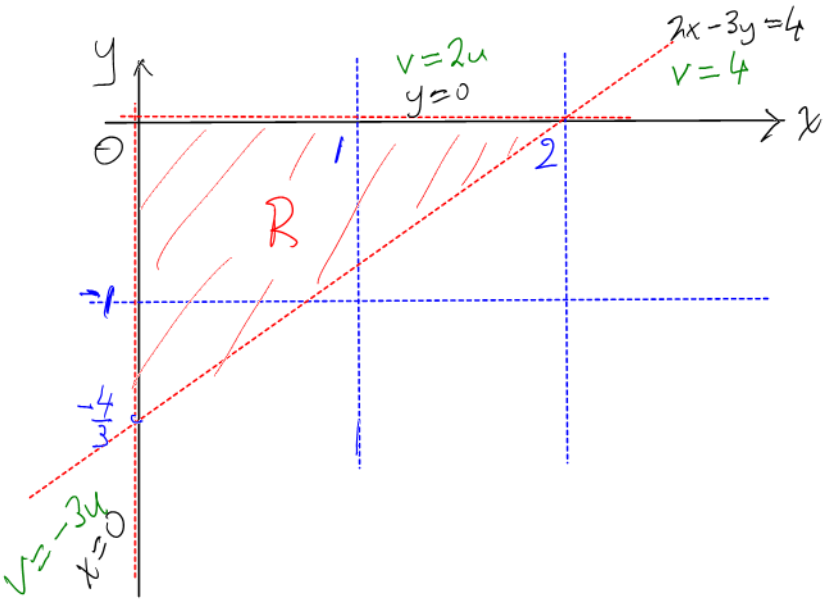
$$J_{\Phi}(x,y) = \det\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \det\begin{pmatrix} 1 & -1 \\ 1 & +1 \end{pmatrix} = 2$$

Const Indep of (x,y)

Evaluate $\iint_R (2x-3y)^2(x+y)^2 \, dx\,dy$

R is the triangle bounded by the positive x-axis, negative y-axis, and line $2x-3y=4$.

Use the change of variables $u = x+y, \quad v = 2x-3y$.



Φ $\quad u = x+y$, $\quad v = 2x-3y$

We could evaluate integral as

$$\iint_R (2x-3y)^2(x+y)^2 \, dx\,dy$$

$$= \int_{x=0}^{x=2} \int_{y=\frac{2x-4}{3}}^{y=0} (2x-3y)^2(x+y)^2 \, dy\,dx$$

expand.

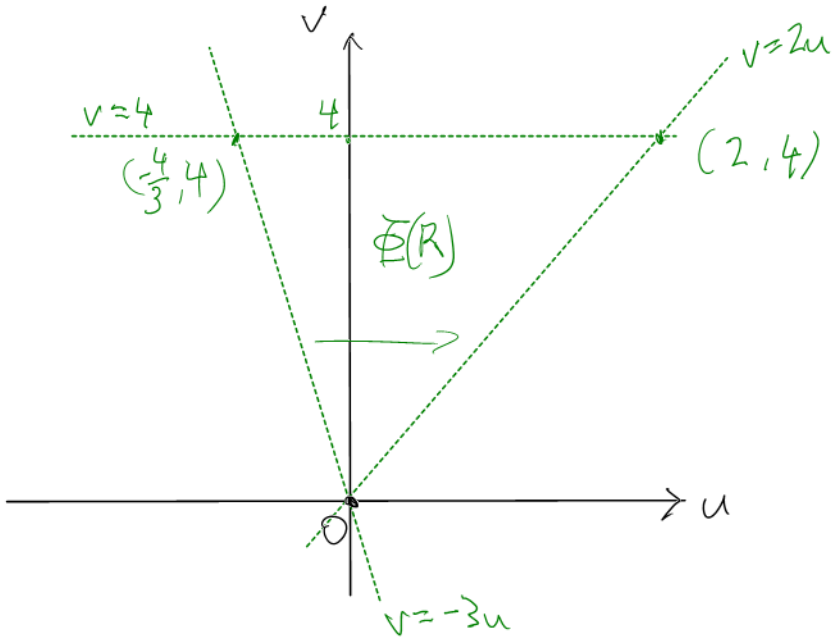
Or: use change of var formula:

$$\iint_{\Phi(R)} f(u,v)\,du\,dv = \iint_R f\big(\Phi(x,y)\big)\left|J_\Phi(x,y)\right| dx\,dy$$

+1

Take $f(u,v) = v^2u^2$

so $f\big(\Phi(x,y)\big) = f\big(\overset{u}{x+y}, \overset{v}{2x-3y}\big) = (2x-3y)^2(x+y)^2$



and $J_\Phi(x,y) = \det\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \det\begin{pmatrix} 1 & 1 \\ 2 & -3 \end{pmatrix} = -5$ const.

So we get (sway LHS, RHS)

$$\iint_R (2x-3y)^2(x+y)^2 \cdot |-5|^{\pm 1}\, dx\,dy = \iint_{\Phi(R)} v^2 u^2 \, du\,dv$$

$$= \int_{v=0}^{v=4} \left(\int_{u=-\frac{v}{3}}^{u=\frac{v}{2}} v^2 u^2 \, du \right) dv$$

$$= \int_{v=0}^{v=4} v^2 \left[\frac{u^3}{3}\right]_{u=-\frac{v}{3}}^{u=\frac{v}{2}} dv = \frac{27+8}{3\cdot 8\cdot 27} \int_{v=0}^{v=4} v^5 \, dv$$

$\left[\frac{u^3}{3}\right]_{u=-\frac{v}{3}}^{u=\frac{v}{2}} = \frac{1}{3}\left(\frac{v^3}{8} + \frac{v^3}{27}\right)$

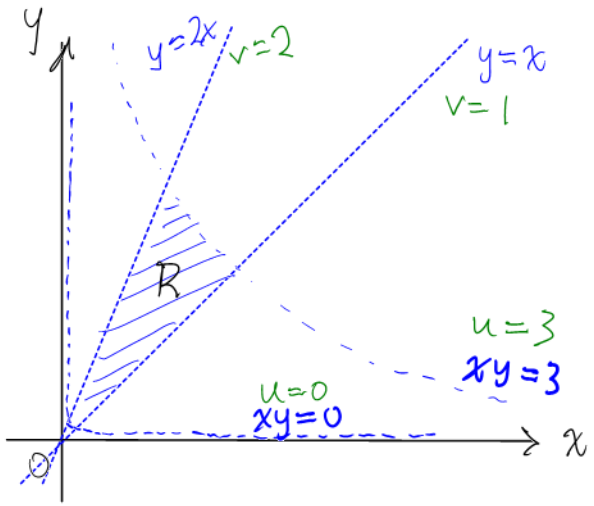
$\left[\frac{v^6}{6}\right]_{v=0}^{v=4} = \frac{4^6}{6}$

So $\iint_R (2x-3y)^2(x+y)^2 \, dx\,dy = \frac{4^4 \cdot 7 \cdot 5}{243}$

$= \cdots = \frac{4^4 \cdot 7 \cdot 5}{243}$

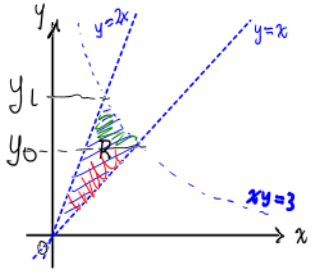
Evaluate $\iint_R (x^2 + y^2)\,dx\,dy$

R is the finite region in the first quadrant bounded by the lines $y = x$ and $y = 2x$, and a portion of the hyperbola $xy = 3$.



Use Ch 1 Var:

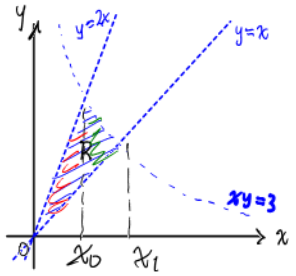
$$\Phi(x,y) = \begin{pmatrix} u(x,y) := xy \\ v(x,y) := \frac{y}{x} \end{pmatrix}$$



$$\iint - \,dx\,dy$$

$$= \int_{y=0}^{y=y_0} \int_{x=y/2}^{x=y} - \,dx\,dy$$

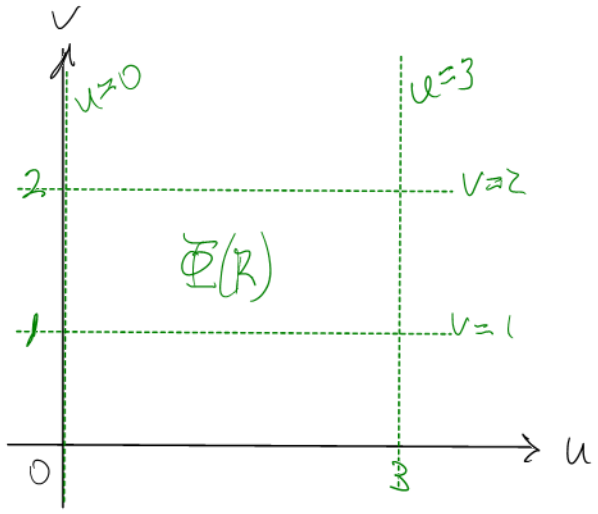
$$+ \int_{y=y_0}^{y=y_1} \int_{x=y/2}^{x=3/y} - \,dx\,dy$$



$$\iint - \,dy\,dx$$

$$= \int_{x=0}^{x=x_0} \int_{y=x}^{y=2x} - \,dy\,dx$$

$$+ \int_{x=x_0}^{x=x_1} \int_{y=x}^{y=3/x} - \,dy\,dx$$



$$(x^2+y^2) = \frac{u}{v} + uv$$

$\uparrow$ $\uparrow$

x^2 y^2

i.e. take $f(u,v)$ to be $\frac{u}{v} + uv$

so that $f(\Phi(x,y)) = x^2+y^2$.

$$J_\Phi(x,y) = \det\begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} = \det\begin{pmatrix} y & x \\ -\frac{y}{x^2} & \frac{1}{x} \end{pmatrix} = \frac{y}{x} - \left(-\frac{y}{x}\right) = \frac{2y}{x} = 2v$$

Ch of Var formula says:

$$\iint_R \underbrace{(x^2+y^2)}_{f(\Phi(x,y))}\,dx\,dy = \iint_{\Phi(R)} \underbrace{\left(\frac{u}{v}+uv\right)}_{f(u,v)} \left|\underbrace{J_\Phi(\Phi^{-1}(u,v))}_{2v}\right|^{-1} du\,dv$$

$$= \iint_{\Phi(R)} \left(\frac{u}{2v^2} + \frac{u}{2}\right) du\,dv = \int_{v=1}^{v=2}\int_{u=0}^{u=3} \left(\frac{u}{2v^2}+\frac{u}{2}\right) du\,dv$$

$$= \int_{v=1}^{v=2} \frac{1}{2v^2}\left[\frac{u^2}{2}\right]_{u=0}^{u=3} + \frac{1}{2}\left[\frac{u^2}{2}\right]_{u=0}^{u=3} dv = \cdots = \frac{9}{4}\left[-\frac{1}{v}+v\right]_{v=1}^{v=2}$$

$$= \frac{27}{8} \#$$